1. USB Webcam Connection, Driver Installation and Testing

Aim

To connect a USB webcam to a computer, install the required drivers, and verify that the webcam is working properly by using the default Windows Camera application and capturing a photograph.

Requirements

- Computer or Laptop
- USB Webcam
- USB Port
- Windows Operating System
- Webcam Driver, if required
- Windows Camera Application

Introduction

A webcam is an input device used to capture video and images. USB webcams are generally detected automatically by Windows when they are connected to the computer. If Windows does not automatically install the required driver, the driver can be installed manually.

Testing the webcam after installation is important because it confirms that the device, driver and camera application are functioning correctly.

Procedure

1. Turn on the computer and wait until Windows has completely started.
2. Take the USB webcam and inspect its USB connector.
3. Connect the webcam to an available USB port on the computer.
4. Wait for Windows to detect the newly connected hardware.
5. Windows may automatically install the required webcam driver.
6. Open **Device Manager** by right-clicking the Start button and selecting **Device Manager**.
7. Locate the **Cameras** section.
8. Expand the section and check whether the connected webcam is listed.
9. If the webcam is not detected or a warning icon is displayed, check and install the appropriate driver.
10. Restart the computer if required after driver installation.
11. Open the Windows **Camera** application from the Start menu.
12. If Windows asks for camera permission, select the option to allow camera access.
13. Check the live camera preview.
14. Move the webcam or change its position to verify that the live video changes correctly.

15. Adjust the webcam if necessary so that the captured image is clear.
16. Capture a test photograph using the Camera application.
17. Open the saved photograph and verify that the image has been captured successfully.
18. Close the Camera application after completing the test.

Observation

The webcam was detected by the Windows operating system and appeared under the Cameras section of Device Manager. The Camera application displayed live video and successfully captured a photograph.

Result

The USB webcam was successfully connected, the required driver was installed/verified, and the webcam was tested successfully using the Windows Camera application.

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2. Local Printer Setup, Driver Installation and Test Page

Aim

To set up a local printer using USB or a network connection, install the correct printer driver, and print a test page to verify successful installation.

Requirements

- Computer or Laptop
- Printer
- USB Cable or Network Connection
- Printer Driver
- Printer Paper
- Power Supply

Introduction

A printer is an output device used to produce documents, images and other information on paper. Before a printer can be used, it must be properly connected and configured. Windows normally detects many printers automatically, but some printers require a specific driver.

Printing a test page is a simple way to confirm that the printer has been installed correctly.

Procedure

1. Place the printer near the computer.
2. Connect the printer to a suitable power supply.
3. Turn on the printer.
4. Make sure the printer has paper loaded.

5. For a USB printer, connect the USB cable between the printer and computer.
6. For a network printer, connect the printer to the appropriate Wi-Fi or network.
7. Open **Windows Settings**.
8. Go to **Bluetooth & devices**.
9. Select **Printers & scanners**.
10. Click **Add device**.
11. Allow Windows to search for available printers.
12. Select the required printer from the list.
13. Follow the instructions displayed by Windows.
14. If Windows cannot automatically install the driver, install the correct driver provided by the printer manufacturer.
15. After installation, return to **Printers & scanners**.
16. Select the installed printer.
17. Check that the printer is shown as available/ready.
18. Open the printer properties or management options.
19. Select **Print Test Page**.
20. Wait for the printer to print the test page.
21. Check the printed page for proper text, alignment and print quality.
22. If the page prints correctly, the printer installation can be considered successful.

Observation

The printer was detected by Windows and appeared under Printers & scanners. The required driver was installed and the printer successfully printed a test page.

Result

The local printer was successfully connected and configured. The successful printing of the test page confirmed that the printer installation was working correctly.

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3. Troubleshooting a "Printer Not Detected" Problem

Aim

To simulate a printer connectivity problem by disconnecting the printer's USB cable and troubleshoot the problem using Windows Device Manager.

Requirements

- Computer or Laptop

- Installed USB Printer
- USB Cable
- Windows Operating System
- Device Manager

Introduction

Sometimes Windows may fail to detect a printer because of a loose USB connection, damaged cable, incorrect driver, USB port problem or a temporary hardware detection issue.

Windows Device Manager provides useful information about connected hardware and can be used to identify devices with driver or hardware problems.

Problem Simulation

The printer is first connected and working correctly. The USB cable is then disconnected from the computer to simulate a connectivity problem. After that, the printer is reconnected and Windows Device Manager is used to detect the hardware again.

Procedure

1. First confirm that the printer is working correctly.
2. Disconnect the printer's USB cable from the computer.
3. Open **Settings → Bluetooth & devices → Printers & scanners**.
4. Check whether the printer is unavailable or no longer detected.
5. Open **Device Manager**.
6. Expand the **Printers** or **Print queues** section.
7. Also check **Universal Serial Bus controllers**.
8. Look for devices showing a yellow warning icon.
9. A yellow warning icon may indicate a driver or hardware detection problem.
10. Reconnect the printer's USB cable securely.
11. Wait for Windows to detect the printer again.
12. In Device Manager, click **Action**.
13. Select **Scan for hardware changes**.
14. Windows will scan the computer for newly connected hardware.
15. Wait for the printer to appear again.
16. Open **Printers & scanners** and verify that the printer is detected.
17. If the printer still does not work, right-click the affected device in Device Manager.
18. Select **Update driver**.

19. Follow the instructions to search for and install the appropriate driver.
20. If necessary, reinstall the printer driver.
21. Restart the computer if Windows requests a restart.
22. Print a test page after reconnecting the printer.

Troubleshooting Checks

- Check whether the USB cable is connected properly.
- Try another USB port.
- Check the printer's power supply.
- Check Device Manager for yellow warning icons.
- Use **Scan for hardware changes** after reconnecting the printer.
- Update or reinstall the printer driver if required.
- Restart Windows and test the printer again.

Observation

After the USB cable was disconnected, the printer was unavailable to the computer. After reconnecting the cable and selecting **Scan for hardware changes**, Windows detected the hardware again.

Result

The simulated "printer not detected" problem was successfully identified and resolved using the USB connection, Device Manager and the **Scan for hardware changes** feature.

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4. Troubleshooting Webcam Video Not Displaying in Zoom

Aim

To troubleshoot and resolve a problem where the webcam works correctly in the default Camera application but does not display video in Zoom.

Requirements

- Computer or Laptop
- Webcam
- Windows Operating System
- Zoom Application
- Camera Application
- Internet Connection

Problem Description

A webcam may work normally in the Windows Camera application but fail to display video in Zoom. Common reasons include another application using the webcam, disabled Windows camera permissions, incorrect camera selection in Zoom or a webcam driver problem.

Procedure

Step 1: Test the Webcam

1. Open the Windows **Camera** application.
2. Check whether the webcam displays live video.
3. If the webcam works correctly, close the Camera application.
4. Make sure the Camera application is completely closed.

Step 2: Close Other Applications

1. Check whether another application is using the webcam.
2. Close applications such as Microsoft Teams, Skype, Google Meet or other video applications.
3. Close browser tabs that may be accessing the webcam.
4. Restart Zoom after closing these applications.

Step 3: Check Windows Camera Permissions

1. Open **Windows Settings**.
2. Go to **Privacy & security**.
3. Select **Camera**.
4. Make sure **Camera access** is enabled.
5. Make sure applications are allowed to access the camera.
6. Ensure that desktop applications are permitted to access the camera where applicable.

Step 4: Check Zoom Camera Settings

1. Open the Zoom application.
2. Open **Settings**.
3. Select **Video** or **Video & effects**.
4. Locate the **Camera** option.
5. Check the list of available cameras.
6. Select the correct webcam.
7. Check the video preview.
8. If the preview displays live video, the webcam is working correctly in Zoom.

Step 5: Check Device Manager

1. Right-click the Windows Start button.

2. Open **Device Manager**.
3. Expand **Cameras**.
4. Locate the webcam.
5. Check whether there is a yellow warning icon.
6. If a driver problem is shown, right-click the webcam.
7. Select **Update driver**.
8. Install the appropriate driver if an update is available.
9. Restart the computer if required.

Step 6: Test Zoom Again

1. Open Zoom again.
2. Open the video settings.
3. Select the correct webcam.
4. Check the camera preview.
5. Start a test meeting if necessary.
6. Turn on the camera.
7. Verify that live video is displayed.

Common Causes and Solutions

Problem	Solution
Another application is using the webcam	Close other camera/video applications
Camera permission is disabled	Enable camera access in Windows Privacy settings
Wrong camera selected	Select the correct webcam in Zoom
Webcam driver problem	Update or reinstall the webcam driver
USB connection problem	Reconnect webcam or try another USB port
Temporary application problem	Close and restart Zoom

Observation

The webcam worked correctly in the Windows Camera application. After closing other applications, checking Windows camera permissions and selecting the correct webcam in Zoom, the camera preview was displayed correctly.

Result

The webcam problem in Zoom was successfully resolved. The webcam worked correctly in both the Windows Camera application and Zoom.

Overall Conclusion

This assignment covered four important practical activities related to computer peripherals and troubleshooting.

First, a USB webcam was connected to the computer, its driver was checked, and its operation was verified using the Windows Camera application.

Second, a local printer was connected and configured. The appropriate driver was installed and a test page was printed to confirm successful installation.

Third, a printer connectivity problem was simulated by disconnecting the USB cable. Windows Device Manager was used to identify the problem, and the **Scan for hardware changes** option was used after reconnecting the printer.

Finally, a webcam problem in Zoom was investigated. Camera permissions, other applications using the webcam, Zoom's camera selection and webcam drivers were checked. After troubleshooting, the webcam was successfully used in Zoom.

Learning Outcomes

After completing this practical assignment, the student will be able to:

1. Connect and test USB computer peripherals.
2. Install and verify hardware drivers.
3. Configure a local printer.
4. Print a printer test page.
5. Use Windows Device Manager for hardware troubleshooting.
6. Identify common printer connectivity problems.
7. Use the Scan for hardware changes feature.
8. Check Windows privacy and camera permissions.
9. Configure webcam settings in Zoom.
10. Troubleshoot common webcam and printer problems.